

AEDR QCEW Engineering Audit

Dataset

Quarterly Census of Employment and Wages (QCEW) – Annual Single File

Date

August 7, 2026

Observation 001

Description

During the initial import of the Annual QCEW files (1990–2025) into SQLite, a data type inconsistency was observed.

The BLS source files contain an identical column layout across all years (38 columns). However, DB Browser for SQLite automatically inferred different SQLite data types for the `industry_code` field.

Observed behavior:

- **1990–2003:** `industry_code` imported as **TEXT**
- **2004–2025:** `industry_code` imported as **INTEGER**

No evidence has been found that the underlying BLS schema changed. This appears to be a consequence of SQLite's automatic type inference during CSV import.

Impact

This does **not** represent a source data inconsistency.

It is an import behavior introduced by the SQL import process.

AEDR Standard

For the American Economic Data Repository (AEDR), `industry_code` will be standardized as **TEXT** across all years.

Although many industry codes appear numeric, they function as identifiers rather than values used in mathematical calculations. Treating them as text preserves formatting consistency and avoids potential issues with future revisions or special code values.

Action Required

During Phase 2 (AEDR Engineering):

- Standardize industry code to **TEXT**
 - Verify any additional identifier fields that SQLite may infer differently across years
 - Document all normalization decisions in this engineering log
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Observation 002 – Significant Row Count Increase Beginning in 2001

The annual QCEW single file increases from approximately 1.0 million records (1990–2000) to approximately 3.6 million records beginning in 2001.

This likely reflects a change in the published dataset rather than an import error.

Action: Verify BLS documentation to determine the cause before normalization.

Observation 003 – Aggregation Levels

Status: Verified

The annual QCEW files for 1990 and 2001 contain the same number of aggregation levels.

- 1990: 44
- 2001: 44

Conclusion:

The increase in total row count beginning in 2001 is **not** the result of additional aggregation levels.

Observation 004 — Ownership Code Comparison

Description

As part of the investigation into the significant increase in QCEW annual row counts beginning in 2001, the distinct values of own_code were compared between 1990 and 2001.

Results

- **1990:** 7 distinct own_code values
- **2001:** 7 distinct own_code values

Conclusion

The increase in QCEW row counts beginning in 2001 is **not explained by an increase in the number of ownership categories.**

The ownership code structure appears consistent between the two comparison years.

Status

- ☒ Verified
 - ☒ No normalization issue identified
 - ☐ Further action required
-

Observation 005 – Establishment Size Code Comparison

Description

As part of the investigation into the significant increase in QCEW annual row counts beginning in 2001, the distinct values of size_code were compared between 1990 and 2001.

The objective was to determine whether additional establishment size classifications were introduced beginning in 2001.

Results

- **1990:** 1 distinct size_code value
 - **2001:** 1 distinct size_code value
-

Conclusion

The increase in QCEW annual row counts beginning in 2001 is **not explained by changes to the establishment size classifications.**

The size_code structure appears consistent between the two comparison years.

Status

- ☒ Verified
 - ☒ No normalization issue identified
 - ☒ Further action required
-

Observation 006 – Geographic Coverage Comparison

Description

As part of the investigation into the significant increase in QCEW annual row counts beginning in 2001, the number of distinct area_fips values was compared between 1990 and 2001.

The objective was to determine whether expanded geographic coverage accounted for the increase in records.

Results

- **1990:** 4,343 distinct area_fips values
- **2001:** 4,400 distinct area_fips values

This represents an increase of **57 geographic identifiers**.

Conclusion

While the annual QCEW files include a modest increase in geographic coverage beginning in 2001, the addition of 57 area_fips values is **not sufficient to explain** the increase in annual row counts from approximately 1.0 million to 3.6 million records.

The primary cause of the increase lies elsewhere in the dataset.

Status

- ☒ Verified
 - ☒ Minor structural difference identified
 - ☒ Further investigation required
-

Observation 007 – Industry Code Comparison

Description

As part of the investigation into the significant increase in QCEW annual row counts beginning in 2001, the number of distinct industry_code values was compared between 1990 and 2001.

The objective was to determine whether additional industry classifications accounted for the increase in records.

Results

- **1990:** 2,370 distinct industry_code values
- **2001:** 2,376 distinct industry_code values

This represents an increase of **6 industry codes**.

Conclusion

The increase in QCEW annual row counts beginning in 2001 is **not explained by changes in the number of industry classifications**.

The industry classification structure remains substantially consistent between the two comparison years.

Status

- ☒ Verified
- ☒ Minor structural difference identified
- ☒ Further investigation required

Observation 008 — Record Combination Comparison

Description

As part of the investigation into the significant increase in QCEW annual row counts beginning in 2001, unique combinations of the primary categorical identifiers were compared between 1990 and 2001.

The fields evaluated were:

- area_fips
- own_code
- industry_code
- agglvl_code
- size_code

Results

- **1990:** 930,835 unique combinations
- **1990 total rows:** 930,835
- **2001:** 3,579,633 unique combinations
- **2001 total rows:** 3,579,633

Conclusion

Every record in both comparison years represents a unique combination of the tested identifier fields.

The increase in QCEW annual row counts beginning in 2001 is therefore **not caused by duplicate records**.

Instead, the 2001 file contains substantially more combinations of existing geographic, industry, ownership, aggregation-level, and size identifiers.

Further investigation is required to determine **which part of the reporting structure accounts for the additional combinations**.

Status

- ☒ Verified
- ☒ Duplicate records ruled out
- ☒ Further investigation required

Observation 009 — 2001 NAICS Transition and Expansion of Detailed Industry Records

Description

The distribution of QCEW records by agglvl_code was compared between 1990 and 2001 to determine where the significant increase in annual row counts occurred.

Results

The largest increases occurred in detailed industry aggregation levels, particularly:

- 78 — County, 6-digit industry by ownership: **+853,874**
- 77 — County, 5-digit industry by ownership: **+703,849**
- 76 — County, 4-digit industry by ownership: **+404,317**
- 75 — County, 3-digit industry by ownership: **+174,289**
- 48 — Statewide, 6-digit industry by ownership: **+180,745**
- 47 — Statewide, 5-digit industry by ownership: **+124,516**

BLS documentation confirms that **2001 was the first official QCEW reference year produced using NAICS**, replacing the SIC classification system used through 2000.

BLS also provides reconstructed NAICS-based QCEW files for 1990–2000. These historical files were reconstructed from SIC-classified data and do not contain the same level of detailed industry coverage available in the native NAICS files beginning in 2001.

Conclusion

The sharp increase in QCEW annual row counts beginning in 2001 is primarily associated with a substantial expansion in detailed industry-by-geography reporting, particularly county and state records at the 3-, 4-, 5-, and 6-digit industry levels.

The increase does **not** appear to be an import error or duplicate-record issue.

Instead, it reflects the transition to native NAICS-based QCEW reporting beginning in 2001 and the greater industry detail available in those records.

AEDR Impact

The 1990–2000 and 2001-forward files can remain in the same QCEW dataset, but the historical distinction must be documented.

Detailed industry-level comparisons spanning the 2000–2001 boundary must be treated cautiously because the underlying classification/reporting structure changed.

Status

- ☒ Cause identified
- ☒ Structural change documented
- ☒ Import error ruled out
- ☒ Normalization note required

Now that **Observations 001–009** have established that the files imported correctly and that **the major structural anomaly is legitimate**, the next logical step is to stop examining structure and start examining **data quality**.

I would make **Observation 010 — Critical Field Integrity Audit**.

Rather than checking every one of the 38 columns immediately, start with the fields that define what a QCEW observation actually *is*:

area_fips, own_code, industry_code, agglvl_code, size_code, and year.

We want to answer one simple question:

Does every record have a complete, valid identifying structure?

Specifically, across **all 1990–2025 tables**, check for **NULL or blank values** in those fields. This is different from the work in Observations 003–008: there we compared the *structure and distinct values* between 1990 and 2001. We have **not yet established whether records throughout the complete 36-year series contain missing identifiers**.

I would **not** start by auditing NULLs across all 38 columns. QCEW legitimately contains suppressed/unavailable economic values, so we'd generate a huge amount of noise. First establish that the identifiers are intact.

If that passes, the progression becomes:

Observation 010: Critical identifiers — NULL/blank integrity

- **011:** Year and categorical-code validity
- **012:** Negative/impossible numeric values
- **013:** Suppression/missing-value behavior
- **014:** Business-rule/internal consistency checks

At that point we'd be moving from “**Did BLS give us a structurally coherent dataset?**” to “**Do the actual economic values behave as QCEW says they should?**”

Observation 010 — Critical Identifier NULL/Blank Audit

Description

As part of the QCEW data quality validation process, all annual QCEW Single-File tables from **1990 through 2025** were audited for NULL or blank values in the critical fields used to identify and classify individual records.

The fields evaluated were:

- area_fips

- own_code
- industry_code
- agglvl_code
- size_code
- year
- qtr

The audit evaluated every record in each of the 36 annual tables.

Results

All seven critical identifier fields returned **zero NULL or blank values for every year from 1990 through 2025**.

No missing identifiers were detected in any annual table.

The audit also confirmed the previously observed annual row-count pattern, including the documented increase beginning in 2001.

The 2025 file contains **3,088,320 records**, compared with **3,664,909 records in 2024**. This decrease was observed during the audit but is **not considered evidence of a data-quality problem based on this test alone**. Any change in QCEW publication coverage or reporting granularity affecting 2025 should be documented separately.

Conclusion

The QCEW Annual Single-File dataset passes the critical identifier completeness audit.

Every record from 1990 through 2025 contains populated values for all seven fields necessary to identify and classify the observation.

No missing critical identifiers or apparent identifier-related import failures were detected.

AEDR Impact

No corrective action is required for missing critical identifiers.

The 2025 row-count decrease should be treated separately as a potential source-publication or structural change rather than as a NULL-integrity failure.

Status

- ☒ Verified
 - ☒ 1990–2025 audited
 - ☒ No missing critical identifiers detected
 - ☒ No identifier completeness normalization required
 - ☐ 2025 row-count change to be documented separately
-

Observation 011 — Core Value Validity Audit

Description

As part of the QCEW data quality validation process, all annual QCEW Single-File tables from **1990 through 2025** were audited for invalid temporal values, negative categorical codes, and impossible negative values in core employment and wage measures.

The following conditions were tested:

- year does not match the corresponding annual file
 - qtr is not equal to A
 - Negative own_code
 - Negative agglvl_code
 - Negative size_code
 - Negative annual_avg_estabs
 - Negative annual_avg_emplvl
 - Negative total_annual_wages
 - Negative annual_avg_wkly_wage
 - Negative avg_annual_pay
-

Results

All tested conditions returned **zero invalid records for every annual file from 1990 through 2025**.

No mismatched years, invalid annual-period indicators, negative categorical codes, or negative values in the tested establishment, employment, and wage measures were detected.

Conclusion

The QCEW Annual Single-File dataset passes the core value validity audit.

The temporal identifiers are internally consistent with their source files, and the tested core economic measures contain no impossible negative values.

AEDR Impact

No corrective action or normalization is required as a result of this audit.

Negative values in fields explicitly representing changes or percentage changes were intentionally excluded from this test because negative changes are economically valid.

Status

- ☒ Verified
- ☒ 1990–2025 audited
- ☒ Year/file consistency verified
- ☒ Annual-period indicator verified
- ☒ No negative core level values detected
- ☒ No corrective action required

Next, I would move to **Observation 012 — Duplicate Record Audit across 1990–2025**. We proved 1990 and 2001 had no duplicates during the earlier investigation, but we haven't yet tested all 36 years for duplicate identifier combinations.

Observation 012 — Duplicate Record Audit

Description

All QCEW Annual Single-File tables from **1990 through 2025** were audited for duplicate records using the combination of the primary categorical identifier fields:

- area_fips
- own_code

- industry_code
- agglvl_code
- size_code

Records were grouped by these five fields, and any identifier combination occurring more than once was classified as a duplicate group.

Results

All 36 annual tables returned **zero records belonging to duplicate identifier groups**.

For every year from 1990 through 2025, the number of unique identifier combinations equals the total number of records in the corresponding annual table.

Conclusion

Every QCEW record from 1990 through 2025 represents a unique combination of the five tested categorical identifiers.

No duplicate records were detected.

This extends the earlier duplicate analysis performed for 1990 and 2001 to the **entire QCEW annual dataset**.

AEDR Impact

No deduplication or corrective action is required.

The combination of area_fips, own_code, industry_code, agglvl_code, and size_code functions as a unique record key within each annual QCEW file.

Status

- ☒ Verified
- ☒ 1990–2025 audited
- ☒ Zero duplicate identifier combinations detected
- ☒ No deduplication required

At this point we've now established **completeness, basic value validity, and uniqueness** across the whole dataset.

The next logical audit is different and more meaningful: **business-rule/internal consistency validation**—for example, checking whether BLS's reported avg_annual_pay and annual_avg_wkly_wage are internally consistent with the underlying wage/employment figures, accounting for BLS definitions and rounding. That's where we start testing whether the numbers relate to each other the way they're supposed to, rather than just checking whether individual values are valid.

Observation 013 — Average Annual Pay / Weekly Wage Internal Consistency

Description

As part of QCEW business-rule validation, the relationship between avg_annual_pay and annual_avg_wkly_wage was tested across all annual QCEW Single-File tables from **1990 through 2025**.

The expected relationship is:

$$\text{Average Annual Pay} \approx \text{Average Weekly Wage} \times 52$$

Because annual_avg_wkly_wage is reported in whole dollars, multiplication of the rounded weekly value by 52 can produce a difference of up to approximately **\$26** relative to the reported annual-pay value.

The audit therefore tested every record for:

$$\text{ABS}(\text{avg_annual_pay} - (\text{annual_avg_wkly_wage} * 52)) > 26$$

Any record exceeding \$26 would have been flagged for further investigation.

Results

All annual tables from **1990 through 2025** returned **zero records exceeding the \$26 rounding tolerance**.

Preliminary testing of 1990–1992 also found a maximum observed difference of exactly **\$26**, with an average absolute difference of approximately **\$13**.

These results are consistent with the expected effect of whole-dollar rounding of the reported weekly-wage measure.

Conclusion

The reported avg_annual_pay and annual_avg_wkly_wage fields are internally consistent throughout the QCEW dataset within the expected rounding tolerance.

No anomalous discrepancies were detected.

AEDR Impact

No corrective action or normalization is required.

The observed differences between annual pay and annualized weekly wages are consistent with expected rounding rather than data-quality or import errors.

Status

- ☒ Verified
- ☒ 1990–2025 audited
- ☒ Zero records outside expected rounding tolerance
- ☒ Internal wage/pay relationship validated
- ☒ No corrective action required

Next I'd test an even more fundamental QCEW relationship: **whether avg_annual_pay agrees with total_annual_wages ÷ annual_avg_emplvl**. That lets us validate the reported pay measure against its underlying wage and employment fields rather than against another derived pay measure.

Observation 014 — Average Annual Pay Reconstruction and Annual Employment Rounding

Description

As part of QCEW business-rule validation, the published avg_annual_pay field was compared with a reconstructed value calculated as:

Total Annual Wages ÷ Published Annual Average Employment

The objective was to determine whether the relationship among total_annual_wages, annual_avg_emplvl, and avg_annual_pay was internally consistent.

Validation Scope

The calculation was evaluated across all annual QCEW files from **1990 through 2025**.

For each year, the following were calculated:

- Number of records with $\text{annual_avg_emplvl} > 0$
- Maximum absolute difference between reported and reconstructed annual pay
- Average absolute difference between reported and reconstructed annual pay

Differences were present throughout the dataset.

Average absolute differences increased over time, from approximately **\$95.40 in 1990** to **\$604.34 in 2025**. Maximum differences were substantially larger for individual records, with the largest observed difference occurring in **2021 at \$1,268,199**.

Investigation

Records producing large discrepancies were manually inspected.

The largest differences were concentrated among highly granular observations with very small published annual average employment levels, frequently $\text{annual_avg_emplvl} = 1$.

A 1990 record, for example, reported:

$\text{annual_avg_emplvl} = 3$

$\text{total_annual_wages} = \$852,928$

$\text{avg_annual_pay} = \$341,171$

Using the published employment value produces:

$\$852,928 \div 3 = \$284,309.33$

However, solving for the employment level implied by BLS's reported annual pay produces approximately:

$\$852,928 \div \$341,171 \approx 2.5$

The same pattern was observed in other records, including cases where the reported annual pay implied underlying annual average employment of **0.5** despite a published annual_avg_emplvl of **1**.

Extreme-Value Verification

The largest discrepancy identified across the entire 1990–2025 audit occurred in 2021.

The record reported:

area_fips = 31999

own_code = 1

industry_code = 921130

agglvl_code = 78

annual_avg_emplvl = 1

total_annual_wages = \$6,340,995

avg_annual_pay = \$5,072,796

Using the published employment level produces reconstructed annual pay of **\$6,340,995**, resulting in a difference of **\$1,268,199**.

However:

$\$6,340,995 \div \$5,072,796 \approx 1.25$

The reported annual pay is therefore consistent with an underlying annual average employment level of approximately **1.25**.

Because QCEW annual average employment is derived from monthly employment levels, fractional annual averages can result from averaging the twelve monthly observations even though the annual file publishes employment as a whole-number level.

Conclusion

The differences between reported avg_annual_pay and values reconstructed using published annual_avg_emplvl are consistent with **precision loss associated with the published annual average employment measure**.

The effect becomes especially pronounced for records with very small employment levels because rounding represents a large proportional change in the denominator.

The observed differences therefore **do not indicate corrupted wage or annual-pay data**.

AEDR Impact

The original BLS values should be preserved.

avg_annual_pay should **not** be recalculated or replaced using the published annual_avg_emplvl, because the published employment value may not retain sufficient precision to reproduce the BLS annual-pay calculation exactly.

No corrective normalization is required.

A separate research note has been retained regarding the unusual 2021 federal Public Finance Activities record (area_fips 31999) for possible later investigation; it does **not** prevent QCEW validation from proceeding.

Status

- ☒ Full 1990–2025 dataset tested
- ☒ Maximum and average differences evaluated
- ☒ High-discrepancy records manually investigated
- ☒ Largest 1990–2025 discrepancy investigated
- ☒ Methodological explanation identified
- ☒ Data corruption not indicated
- ☒ Original BLS values to be preserved
- ☒ No corrective action required

Observation 015 — 2025 Row-Count Decline and MSA Publication Change

Description

During the QCEW Annual Single-File audit, the 2025 file was observed to contain substantially fewer records than 2024.

Results

- **2024:** 3,664,909 records
- **2025:** 3,088,320 records
- **Difference:** –576,589 records
- **Decrease:** approximately **15.7%**

The decline was initially flagged for separate investigation because annual QCEW row counts had generally remained near 3.5–3.7 million records since 2001.

Investigation

BLS documentation confirms that beginning with **2025 QCEW data**, the publication of Metropolitan Statistical Area (MSA) data changed.

Detailed industry data for MSAs are no longer published. MSA data are instead limited to the **total covered employment aggregation level**.

Detailed industry data continue to be available at other geographic levels, including counties.

This reduction in published MSA-by-industry combinations provides a documented structural explanation for the lower number of records in the 2025 Annual Single File.

The decline therefore does **not** indicate that the 2025 file was incompletely imported or that records were lost during the AEDR import process.

Conclusion

The reduction from 3,664,909 records in 2024 to 3,088,320 records in 2025 is consistent with a documented **BLS change in QCEW publication granularity**.

The 2025 row-count decline represents a source-publication structural change rather than an AEDR data-quality or import error.

AEDR Impact

The 2025 Annual Single File can remain in the same historical QCEW dataset.

However, analyses involving detailed **MSA-by-industry** observations must account for the change in publication coverage beginning in 2025.

The structural break should be documented in AEDR metadata so users do not interpret the reduction in MSA industry records as an economic decline or missing data.

No corrective normalization is required.

Status

- ☒ 2025 row-count decline investigated
- ☒ BLS publication change identified
- ☒ Source structural change documented
- ☒ Import error ruled out

- ☒ No corrective action required
 - ☒ MSA comparability note required beginning in 2025
-

Observation 016 — QCEW Reference-Code Integrity Validation

Description

Following import of the official BLS QCEW classification/reference tables, a full-series referential-integrity audit was performed against the coded dimensions contained in the QCEW Annual Single Files for **1990–2025**.

The purpose of this validation was to determine whether codes contained in the source observations correspond to valid codes in the official BLS reference material.

Reference Tables

The following QCEW fields were validated against their corresponding BLS reference tables:

QCEW Field BLS Reference

own_code	Ownership Codes and Titles
agglvl_code	Aggregation Level Codes and Titles
size_code	Establishment Size Codes and Titles
area_fips	Area Codes and Titles
industry_code	Industry Codes and Titles

Method

For each coded dimension, values appearing across all 36 annual QCEW tables were compared with the corresponding official BLS reference table using a LEFT JOIN.

The QCEW observations were placed on the left side of the join so that any code appearing in QCEW without a corresponding BLS reference record would be retained as an unmatched observation.

Unmatched records were identified where the corresponding reference-table code was NULL.

For the high-cardinality area_fips and industry_code fields, distinct codes were consolidated across annual tables before comparison to reduce unnecessary processing.

Because industry_code is stored using inconsistent SQLite data types across the historical tables, both sides of the industry comparison were cast to TEXT during validation.

Results

Field	Unmatched Codes	Result
own_code	0	PASS
agglvl_code	0	PASS
size_code	0	PASS
area_fips	0	PASS
industry_code	0	PASS

Every coded value tested across the **1990–2025 QCEW dataset** successfully resolved to an official BLS reference record.

No orphaned ownership, aggregation-level, establishment-size, geographic, or industry codes were identified.

Conclusion

The QCEW Annual Single-File dataset demonstrates complete referential integrity for the five principal coded classification fields tested.

The results provide additional evidence that the annual source files were imported faithfully and that the coded dimensions can be reliably interpreted using official BLS reference material.

The previously identified industry_code storage-type inconsistency does not represent a source-data integrity failure. It remains a normalization issue to be corrected during the subsequent data-engineering phase.

AEDR Impact

The official BLS QCEW reference tables should be retained with the QCEW dataset and used to provide human-readable classifications during construction of the normalized AEDR QCEW dataset.

The following relationships can therefore be used reliably:

area_fips → area_title

own_code → own_title

industry_code → industry_title

agglvl_code → agglvl_title

size_code → size_title

No source records require removal or correction as a result of the reference-code audit.

Status

- ☒ Ownership-code integrity validated
 - ☒ Aggregation-level integrity validated
 - ☒ Establishment-size integrity validated
 - ☒ Geographic-code integrity validated
 - ☒ Industry-code integrity validated
 - ☒ No unmatched reference codes identified
 - ☒ Referential-integrity audit complete
 - ☒ No corrective action required
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QCEW Audit Status: COMPLETE

With Observation 016, I would officially consider the **source-data validation phase complete**.

AEDR QCEW Engineering Change log

Dataset

Quarterly Census of Employment and Wages (QCEW) – Annual Single File

Date

August 8, 2026

Engineering Entry 001 — Creation of Unified Normalized QCEW Annual Table

Date: August 8, 2026

Change Type: Schema normalization / dataset consolidation

Source Tables: 1990_QCEW through 2025_QCEW

Target Table: qcew_annual

Description:

Created a unified normalized QCEW annual table by consolidating all 36 validated annual source tables covering 1990–2025. The original source tables were preserved without modification.

The normalized table retains all 38 source-data fields. Two identifier fields were standardized to TEXT to provide consistent storage across years and preserve their function as categorical identifiers:

- area_fips: standardized to TEXT
- industry_code: standardized to TEXT

All remaining fields retain their appropriate INTEGER, REAL, or TEXT data types based on the validated QCEW schema.

A composite primary key was established using:

year, qtr, area_fips, own_code, industry_code, agglvl_code, and size_code.

The annual source tables were loaded into qcew_annual in controlled batches. Row counts were validated individually by year against the source tables following each load.

Validation Result:

PASS — The completed qcew_annual table contains **100,006,739 rows**, representing **36 annual datasets from 1990 through 2025**. Annual row counts match the validated source tables, identifier normalization was confirmed using SQLite typeof(), and composite-key uniqueness is enforced by the table's primary-key constraint.

Source Data Impact:

None. The validated 1990_QCEW–2025_QCEW source tables were not altered.

Engineering Entry 002 — Creation and Validation of Readable QCEW Reference View

Date: August 8, 2026

Change Type: Reference integration / human-readable analytical view

Source Objects: qcew_annual plus official BLS QCEW reference tables

Target View: qcew_annual_readable

Description:

Created a human-readable SQLite view that preserves the normalized QCEW codes while adding official BLS descriptive titles for the five principal coded dimensions:

- area_fips → area_title
- own_code → own_title
- industry_code → industry_title
- agglvl_code → agglvl_title
- size_code → size_title

The view retains the full normalized QCEW observation structure while exposing readable classification labels for analysis, inspection, and downstream use.

During initial view testing, area_title values returned NULL for numeric geographic codes such as Alabama statewide.

Issue Identified:

The original CSV imports had stored numeric area_fips values as integers. This removed leading zeros before normalization.

For example:

01000 → imported as 1000

Casting the integer value to TEXT during initial normalization produced '1000', which did not match the official BLS reference value '01000'.

The dataset also contains valid alphanumeric QCEW area codes such as C1002, meaning a blanket numeric-formatting transformation could not safely be applied to all geographic identifiers.

Correction:

Numeric-only area_fips values were reformatted to five-character identifiers using:

```
printf('%05d', CAST(area_fips AS INTEGER))
```

within a conditional CASE expression.

Alphanumeric area codes were preserved unchanged.

Representative validation confirmed:

1000 → 01000

1001 → 01001

48000 → 48000

C1002 → C1002

C1010 → C1010

All normalized area identifiers remained stored as TEXT.

Validation Result:

PASS.

The readable view was manually inspected after correction and geographic titles populated successfully.

A final distinct-code reference check returned:

missing_area_titles = 0

missing_ownership_titles = 0

missing_industry_titles = 0

missing_agglvl_titles = 0

missing_size_titles = 0

Every coded dimension represented in the normalized QCEW dataset therefore resolves to an official BLS reference title.

Source Data Impact:

None. Original annual QCEW source tables remain unchanged.

The correction was applied only to the normalized qcew_annual engineering layer.

Status

- ☑ Readable reference view created
- ☑ Geographic leading-zero issue identified
- ☑ Numeric FIPS formatting restored
- ☑ Alphanumeric area codes preserved
- ☑ All five reference joins validated
- ☑ Zero unmatched reference titles
- ☑ Engineering Entry 002 complete

Engineering Entry 003 — Historical NAICS Crosswalk Integration and Final Enriched QCEW View

Date: August 8, 2026

Change Type: Historical classification harmonization / analytical view construction

Source Objects:

qcew_annual, qcew_annual_readable, official BLS QCEW aggregation-level reference data, and historical QCEW NAICS hierarchy crosswalks

Target Objects:

ref_naics_codes
ref_naics_hierarchy
qcew_annual_enriched
qcew_annual_final

Description:

Integrated historical NAICS classification metadata into the normalized QCEW dataset so that industry observations can be interpreted using the appropriate NAICS structure for their historical period.

Five distinct BLS QCEW NAICS hierarchy crosswalks were imported and retained:

- 2002 NAICS — 1,197 rows
- 2007 NAICS — 1,193 rows
- 2012 NAICS — 1,083 rows
- 2017 NAICS — 1,075 rows
- 2022 NAICS — 1,029 rows

Each crosswalk contains 14 fields describing the NAICS hierarchy from the six-digit national-industry level through higher-level BLS classifications, including industry, industry group, subsector, sector, supersector, and domain classifications.

A unified NAICS hierarchy reference view was created to expose the five historical classification vintages through a common structure.

QCEW observations were assigned a NAICS vintage according to observation year:

- 1990–2006 → 2002 NAICS
- 2007–2010 → 2007 NAICS

- 2011–2016 → 2012 NAICS
- 2017–2021 → 2017 NAICS
- 2022–2025 → 2022 NAICS

This preserves the historical classification appropriate to each observation rather than incorrectly interpreting all years using the current 2022 NAICS hierarchy.

The QCEW aggregation-level reference was also integrated to distinguish the geographic and industrial resolution represented by each `agglvl_code`. This allows observations to be interpreted as national, state, county, MSA, and other geographic levels and as total-industry, BLS domain, supersector, or NAICS 2–6 digit industry levels where applicable.

The final analytical view combines:

- Original QCEW measures
- Geographic titles
- Ownership titles
- Industry titles
- Establishment-size titles
- Aggregation-level descriptions
- Assigned NAICS vintage
- NAICS hierarchical classifications

Representative validation of the final view confirmed that a 2025 Alabama statewide private-sector NAICS 11 observation correctly resolved to:

- NAICS vintage: 2022
- Geographic level: State
- Industry level: NAICS Sector (2-digit)
- Domain: Goods-Producing
- Supersector: Natural Resources and Mining
- Sector: Agriculture, Forestry, Fishing and Hunting

Further validation confirmed that aggregation levels correctly distinguish more detailed industry observations, including NAICS subsector, industry-group, industry, and national-industry levels.

Validation Result:

PASS — Historical NAICS crosswalks were successfully incorporated and representative observations resolved to the appropriate NAICS vintage, geographic level, industry level, and hierarchical classifications.

The five source crosswalks were confirmed to be distinct historical NAICS structures rather than duplicate files.

Source Data Impact:

None. The original 1990–2025 annual QCEW source tables and imported BLS reference files remain preserved without modification. Historical classification logic and descriptive enrichment are implemented through reference and analytical views layered above the normalized dataset.

Status

- ☒ Five historical NAICS crosswalks imported
 - ☒ Crosswalk schemas validated
 - ☒ Historical crosswalks confirmed distinct
 - ☒ NAICS vintage assigned by observation year
 - ☒ Aggregation-level metadata integrated
 - ☒ Historical NAICS hierarchy integrated
 - ☒ Geographic and industry resolution exposed
 - ☒ Final enriched analytical view created
 - ☒ Representative records validated
 - ☒ Original source tables preserved
 - ☒ Engineering Entry 003 complete
-

Engineering Entry 004 — Final QCEW Dataset Validation and Sign-Off

Date: August 9, 2026

Change Type: Final dataset validation / production sign-off

Objects Validated:

qcew_annual
qcew_annual_readable
ref_naics_hierarchy
ref_naics_codes
qcew_annual_enriched
qcew_annual_final

Description:

Performed final targeted validation of the completed QCEW engineering stack following normalization, reference integration, historical NAICS enrichment, and creation of the final analytical view.

Validation was conducted using targeted structural and representative analytical tests rather than a full materialization of the approximately 100-million-row final view.

Validation Tests:**1. Database Object Validation — PASS**

Confirmed existence of the normalized master table and all five required reference/analytical views.

2. Historical NAICS Version Boundaries — PASS

Confirmed correct NAICS classification assignments at each historical transition:

2006 → 2002 NAICS

2007 → 2007 NAICS

2010 → 2007 NAICS

2011 → 2012 NAICS

2016 → 2012 NAICS

2017 → 2017 NAICS

2021 → 2017 NAICS

2022 → 2022 NAICS

3. NAICS Hierarchy Integrity — PASS

Validated progressive hierarchical classification using representative utility-industry codes:

22 → Sector (2-digit)

221 → Subsector (3-digit)

2211 → Industry Group (4-digit)

22111 → NAICS Industry (5-digit)

221111 → National Industry (6-digit)

Parent hierarchy fields populated progressively and consistently with the level represented by each industry code.

4. Aggregation-Level Interpretation — PASS

Validated the derived analytical dimensions against official BLS aggregation levels 50–58.

The derived fields correctly distinguish:

- Geographic level
- All-industry versus BLS/NAICS industry detail
- Ownership/coverage aggregation scope

State-level observations correctly progress from total-covered observations through BLS Domain, BLS Supersector, and NAICS 2–6 digit classifications.

5. Reference Completeness — PASS

Tested all **2,854 Alabama statewide observations for 2025**.

Results:

missing_area_title = 0

missing_own_title = 0

missing_industry_title = 0

missing_agglvl_title = 0

missing_size_title = 0

All tested coded dimensions successfully resolved to their official BLS descriptive titles.

6. Master Dataset Structural Validation — PASS

Final canonical master-table check returned:

Total observations: 100,006,739

Distinct years: 36

Minimum year: 1990

Maximum year: 2025

These results match the previously validated annual source-table consolidation.

Final Validation Result:**PASS**

The QCEW dataset is approved as the completed normalized and enriched QCEW component of the American Economic Data Repository.

The canonical normalized dataset remains qcew_annual, while qcew_annual_final provides the primary human-readable analytical layer for downstream research and visualization.

Original annual BLS source tables and reference materials remain preserved without modification.

Status

- ☒ Source-table validation complete
- ☒ Normalized master-table validation complete
- ☒ Reference integration validated
- ☒ Historical NAICS boundaries validated
- ☒ NAICS hierarchy validated
- ☒ Aggregation interpretation validated
- ☒ Representative reference completeness validated
- ☒ Final master-table coverage confirmed
- ☒ QCEW engineering complete
- ☒ QCEW validation complete
- ☒ Approved for AEDR documentation/publication preparation